

UNIT 41: MANUFACTURING SECONDARY MACHINING PROCESSES.

Reflective Report on Traditional Secondary Machining Processes and Personal Engineering Performance.

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EXECUTIVE SUMMARY AND VOCATIONAL CONTEXT.

Scope of the Evaluation.

This evaluative report is a detailed review of the workshop manufacturing activities completed during Assignment 2. Writing from my position as a final-year engineering apprentice within a busy, medium-sized subcontract machine shop, my goal is to provide the workshop supervisor with a clear, honest summary of "lessons learnt."

Instead of simply listing the physical steps taken on the shop floor, this report looks deeply at the reasons behind my technical choices. It analyzes why I chose specific setups, how the chosen cutting parameters affected the work, and how the physical metals reacted during machining.

The report evaluates my performance across three main workshop machines:

- The Manual Centre Lathe (used for turning, facing, and taper work)
- The Vertical Milling Machine (used for squaring, facing, and profiling)
- The Vertical Pillar Drill (used for precise hole drilling, counterboring, and countersinking)

During these practical sessions, I worked with three separate component setups across these three machines, utilizing two distinct raw materials that require completely different machining techniques.

The first material was 6082-T6 aluminium alloy, which is lightweight, soft, and gummy to cut. This was used for two of the components: a symmetrical cylindrical bar turned into a conical plumb bob on the lathe, and a flat, rectangular plate drilled, counterbored, and countersunk on the vertical pillar drill.

The second material was EN8 unalloyed medium-carbon steel, which is significantly tougher, harder, and generates a massive amount of heat during a cut. This was used for the third component: a precision-squared rectangular plate machined entirely on the vertical milling machine.

This document acts as a complete summary of the entire manufacturing run. In the sections below, I will evaluate exactly how well I managed workplace health and safety frameworks under real conditions. I will analyze the specific manual machining and general engineering skills used to hit my drawing tolerances.

Finally, I will reflect on how I handled unexpected floor problems, like tool wear and broken swarf, and discuss how these manual configurations can be altered to make production quicker, more accurate, and more sustainable for the business.

The Apprentice Role and Industrial Standards.

In modern engineering facilities, high-volume production is almost always assigned to automated CNC multi-axis machines. CNC centers are fast, highly repeatable,

and reduce the overall cost per part. However, knowing how to set up and run manual lathes, mills, and drills is still a vital skill for a professional engineer.

In a medium-sized company, manual operations are used every day because they keep the workshop flexible. CNC machines are expensive and time-consuming to set up for minor runs, so manual machining is highly cost-effective for:

- One-off toolroom jobs: Making quick, custom jigs, clamping fixtures, or specialized workholding pieces needed on the main production lines.
- Rapid prototyping: Machining single sample parts for design trials where writing, proving, and simulating a CNC program would take far longer than the actual physical machining run.
- Emergency maintenance and repair: Skimming a worn bearing journal, facing a damaged pump housing, or re-chasing stripped threads on critical plant machinery to get a broken production line back up and running without waiting weeks for replacement parts.

As a final-year apprentice, I am no longer treated as a beginner practicing in a school classroom. I am expected to work to real commercial engineering standards. On the shop floor, quality control is binary: a part is either correct to the drawing or it is scrap. There is no middle ground.

If an apprentice makes a careless mistake, sets the wrong spindle speed, or fails to notice a blunt tool, the consequences are real. It wastes raw material, burns expensive tooling inserts, ties up unbillable machine time, and delays customer deliveries, which directly harms the company's reputation and profits.

This report provides documentable evidence of my growth from a trainee who needs constant instruction to an independent professional on the shop floor. It proves that I can read complex engineering prints to BS 8888 standards, take full responsibility for setting and adjusting machinery safely, and use engineering principles to fix problems independently when things go wrong during a cut.

Analysis of Workspace Objectives And Component Logic.

It is critical to outline exactly what each of the three components demanded and why specific machines were chosen to do the work. The manufacturing plan was split logically by geometry, material characteristics, and tool dynamics.

Component 1: The Conical Plumb Bob (Aluminium 6082-T6 on the Lathe).

The primary challenge with this first component was turning a long, smooth taper that matched the drawing's angular requirements exactly, while achieving a clear

surface texture. Aluminium 6082-T6 was chosen because it is highly machinable and corrosion-resistant, making it perfect for a plumb bob.

However, because it is soft and gummy, it easily marks under high clamping forces and tends to weld itself to the cutting tool tip under friction. The Manual Centre Lathe was the only logical choice for this component, as it allowed me to use the compound slide to dial in precise angles and adjust spindle speeds dynamically to achieve a high-quality finish.

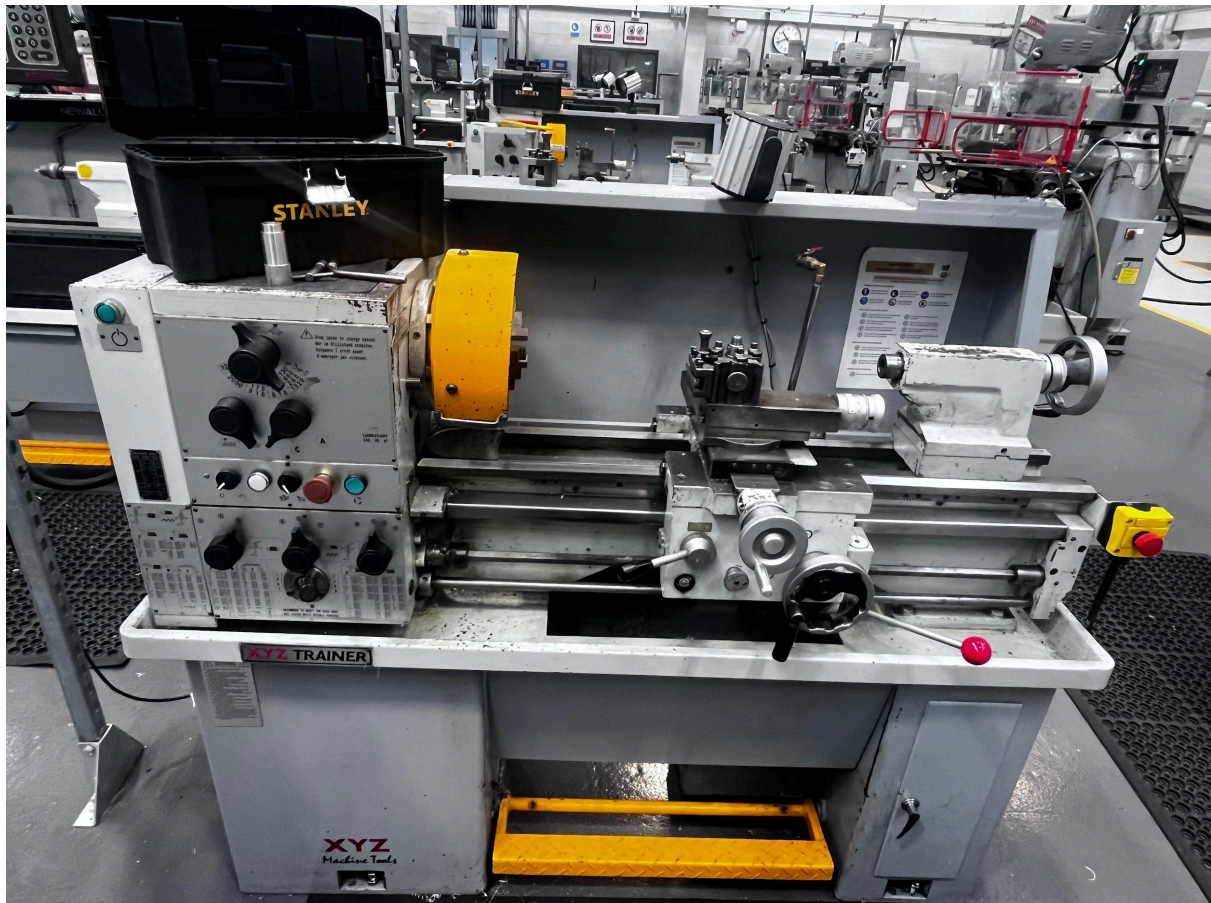


Fig.1 : lathe machine used for component 1. Set- up and safety checks performed before use.

Component 2: The Multi-Feature Plate (Aluminium 6082-T6 on the Pillar Drill).

The second component required processing a flat, rectangular aluminium plate by creating a sequence of complex holes, which included standard parallel holes, multi-diameter counterbores for socket-head cap screws, and countersinks for flush-fitting screws.

The Vertical Pillar Drill was chosen because it allows for direct, manual control and quick tool changes between center drills, twist drills, counterbore cutters, and countersink bits. The core challenge here was depth control and concentricity.

Because aluminium is soft, tools bite rapidly. Preventing the counterbore or countersink from plunging too deep and violating drawing tolerances required careful setting of the machine's mechanical depth stops.

Component 3: The Squared Plate (EN8 Carbon Steel on the Vertical Mill).

This third component demanded flat, square faces and precise external profiling machined from a raw block of tough carbon steel. EN8 was selected because it provides structural strength and rigidity. However, its toughness means it resists cutting forces, creates high tool friction, and produces massive heat spikes that can quickly destroy standard milling cutters if the parameters are wrong.

The Vertical Milling Machine was selected because its rigid vice, heavy table axes, and robust spindle allowed me to ensure that opposing sides were perfectly parallel, while providing the mechanical stability required to drive an end mill or face mill through tough steel without severe tool deflection.



Fig 2 : image of the technical design of the third component and the finished work.

Production Constraints And Material Behaviors.

A major part of working to industrial standards is understanding how different metals behave when a cutting tool hits them. During Assignment 2, I had to constantly adjust my approach depending on which metal was in which machine.

Ductility vs. Hardness.

The aluminium alloy plates and bars behaved unpredictably on both the lathe and the pillar drill. Because it is highly ductile, the metal does not crisp up and snap away easily. Instead, it stretches and deforms before shearing off. If the tool is blunt or the speed is too low, the aluminium will tear rather than cut, leaving a rough surface finish on the lathe or heavy burrs around the edges of the drilled, counterbored, and countersunk holes on the plate.

On the other hand, the EN8 steel block on the milling machine offered high mechanical resistance. Every cut required significant rigidity in the setup. If the workpiece was not clamped down tightly enough in the milling vice, the high cutting forces would cause the part to vibrate, creating a terrible chattering noise that ruins both the tool edge and the surface of the metal.

Thermal Constraints.

Managing heat was an entirely different challenge for each material. Aluminium conducts heat away very quickly, meaning the part itself stays relatively cool, but the tool tip gets hot enough to cause chemical sticking. This is especially true during counterboring and countersinking operations where the tool contact area is large.

Steel acts as a thermal insulator by comparison; the heat builds up right at the cutting edge on the milling machine. Without a proper lubrication and cooling strategy, this intense localized heat can cause the steel to work-harden, making it so hard that the milling cutter cannot bite into it on the next pass.

As a professional engineer, I had to monitor these material behaviors constantly across all three setups, adjust my machine dials to match, and ensure that our production methods remained efficient and safe.

EVALUATION OF HEALTH AND SAFETY MANAGEMENT SYSTEMS.

Statutory Framework Compliance (HSWA and PUWER).

Managing health and safety on the shop floor goes beyond just wearing standard personal protective equipment (PPE). It requires a complete dedication to the legal frameworks that keep a British manufacturing facility running safely.

During the practical phases of Assignment 2, my actions across all three machine setups were governed primarily by two major pieces of UK legislation:

- The Health and Safety at Work etc. Act 1974 (HSWA)
- The Provision and Use of Work Equipment Regulations 1998 (PUWER)

Under Section 7 of the HSWA, I have a clear legal duty as an employee and an apprentice to take reasonable care of my own health and safety, as well as the safety of anyone else in the workshop who could be affected by my actions. I met this standard directly by keeping my immediate machine areas tidy, setting up safety boundaries around my workspace, and strictly following safe operating procedures (SOPs).

Before turning on any machine, I made sure my personal PPE was correct and securely fitted:

- BS EN 166 compliant safety glasses to protect against flying metal chips and spinning swarf.
- BS EN ISO 20345 steel toe-cap boots to safeguard against heavy dropped components like the raw EN8 steel block or the heavy milling fixtures.

- Snug-fitting workshop overalls with the sleeves rolled inward tightly to completely eliminate the risk of the fabric catching on rotating spindles, lathe chucks, or milling arbors.
- No gloves when operating any of the rotating machinery, as gloves represent a severe entanglement hazard if they contact a spinning drill or lathe chuck.

PUWER Regulation 11 focuses on preventing access to dangerous moving parts. I applied this regulation directly during my pre-operational checks on all three machine tools.

On the manual centre lathe, I verified that the chuck guard interlock switch worked perfectly, meaning the spindle could not start spinning while the guard was lifted. On the vertical mill and vertical pillar drill, I checked that the transparent telescopic shields were properly adjusted to completely block the operator zone from flying swarf and splashing coolant.

By checking these systems before releasing the main power isolation switches, I ensured the equipment met required industrial safety baselines before starting any production work.



Fig 3 : safety measures should be adhered to when machining.

Dynamic Risk Assessment and Adaptive Controls.

Pre-printed risk sheets are useful for establishing baseline shop rules, but high-utility engineering demands real-time Dynamic Risk Assessments (DRA) to handle

unexpected process hazards as they happen on the floor. A static document cannot predict every variable of material behavior during a cut, so an apprentice must stay alert and adapt parameters dynamically.

A major lesson was learned regarding dynamic control while machining the rectangular aluminium plate (Component 2) on the vertical pillar drill. The machine was initially set up to run a continuous manual downward feed to drill out the pilot holes.

As the twist drill bit pressed into the soft, flat aluminium plate, it began creating continuous, razor-sharp, ribbon-like swarf. Because aluminium is ductile under high friction, these ribbons did not break away on their own. Instead, they wrapped tightly around the spinning drill chuck, expanding outward like a dangerous mechanical whip right above the workpiece.

Recognizing this immediate entanglement hazard, I carried out a dynamic risk adjustment:

1. I immediately stepped back and hit the stop button on the pillar drill to isolate the machine spindle safely.
2. I cleared the tangled swarf wrap safely using workshop pliers, ensuring my hands stayed clear of the sharp edges.
3. Acting under Regulation 3(1) of the Management of Health and Safety at Work Regulations 1999 (MHSWR), which requires workers to continually review operational risks, I altered my drilling strategy.
4. I changed the machining method from a continuous downward plunge to an aggressive 2 mm peck-drilling cycle.

This technical adjustment fundamentally altered the physics of the cut. By feeding the drill down 2 mm and then retracting it slightly, I cleared the cutting edge and broke the metal chips before they could grow into long ribbons.

The process now produced short, curled, "C-shaped" chips that traveled cleanly up the drill flutes and fell away safely onto the machine table. This proactive change eliminated the risk of entanglement, kept the operation running smoothly, and proved I could maintain an industrial safety standard independently.

Mitigation of Unforeseen Workshop Hazards.

Machining metals across these three setups exposes an engineer to several unexpected thermal, chemical, and physical hazards that must be managed immediately.

Thermal Hazards.

During the roughing phases on the EN8 carbon steel plate on the vertical mill, the

high friction generated a massive amount of heat. Because steel has poor thermal conductivity compared to aluminium, this heat could not escape through the raw block quickly.

Instead, it built up directly at the cutting zone, heating the escaping steel chips to over 300°C. This was confirmed by the chips changing to a distinct blue-straw color as they oxidized in the air.

These hot chips created a fire hazard and could cause severe burns if they hit bare skin. To control this, I lowered the manual feed rate slightly and adjusted the telescopic shielding to deflect the hot chips safely away from my body and into the containment tray.

Chemical Hazards.

To handle the intense heat on both the milling machine and the pillar drill, a continuous flood of 6% water-mix soluble cutting oil emulsion was aimed at the tools. While this fluid was necessary to keep the tools cool, especially when counterboring the aluminium plate to prevent the cutter teeth from clogging, it introduced a chemical hazard.

If cutting fluid is allowed to atomize into a fine mist or splash onto the floor, it can cause respiratory irritation or create a major slip hazard.

To counter this, I regulated the fluid pump valve to keep a steady, low-pressure stream rather than a high-pressure spray, preventing mist formation. I also kept a spill response kit nearby and immediately wiped down any small splashes on the machine housing to keep the floor traction safe.

Physical Structural Hazards.

On the lathe, turning the long taper on the aluminium plumb bob presented a physical projection hazard. At high rotational speeds (1800 rpm), any workpiece material that extends too far out from the back of the chuck can flex and bend violently due to centrifugal forces, a dangerous condition known as "whipping."

To prevent this unforeseen balance issue, I made sure the raw aluminium bar stock was tucked deep into the headstock spindle, leaving only the minimum required material sticking out from the 3-jaw chuck faces.

This smart setup choice kept the lathe perfectly balanced, kept structural vibrations low, and ensured the high-speed turning run was safe from start to finish.

TECHNICAL ANALYSIS OF TRADITIONAL MACHINING SKILLS.

Lathe Operations: Conical Geometry and Surface Integrity.

The main goal when manufacturing Component 1 (the aluminium plumb bob) on the manual centre lathe was executing a clean, uniform taper while maintaining strict control over the surface finish. Because 6082-T6 aluminium is highly ductile and soft, it presents a unique material challenge known as Built-Up Edge (BUE). Under the friction of a cut, the metal can become gummy and weld itself directly onto the rake face of the cutting tool insert. This alters the geometry of the tool tip, causing it to tear the metal rather than shear it cleanly, resulting in a rough, cloudy, and out-of-spec surface texture.

To calculate the correct spindle speed (N) for the roughing and finishing passes, I applied the standard workshop engineering formula.

- Initial Test Cut: Setting the lathe to a conservative 800 rpm caused the tool to rub. The low velocity allowed the hot aluminium chips to stick to the indexable carbide insert tool tip, creating instant BUE and leaving heavy tear marks on the taper face.
- Parameter Optimization: To eliminate the adhesive welding mechanism, I increased the spindle speed to 1800 rpm for the final pass, using a highly positive rake angle insert with a polished face.
- Result: The higher cutting velocity raised the shear-zone temperature just enough to soften the material slice smoothly, eliminating BUE entirely and producing a mirror-like finish that comfortably met the surface roughness rules on the drawing.

To cut the actual conical shape, I swiveled the lathe's manual compound slide to the exact half-angle specified on the blueprint. By locking the carriage firmly to the bedways and feeding the compound slide slowly by hand, I achieved a straight, uniform taper without any steps or gouges.



Fig 4: image of the finished product.

Vertical Milling: Precision Squaring and Workholding.

For Component 3 (the EN8 carbon steel plate), the objective on the vertical milling

machine was to face and square the raw, rough block into a perfectly flat rectangle with parallel sides. Unlike aluminium, EN8 medium-carbon steel is tough and resistant, meaning any instability in the machine setup will cause severe tool deflection, structural vibration, and a destructive phenomenon known as "cutter chatter."

To ensure the block was machined square, workholding preparation was vital:

1. I mounted a Dial Test Indicator (DTI) directly onto the milling machine spindle.
2. I swept the DTI plunger across the face of the fixed vice jaw, adjusting the vice until the dial needle showed a variance of less than ± 0.02 mm across a 100 mm travel span. This step guaranteed that the master datum jaw sat perfectly parallel to the table's X-axis travel.
3. The steel block was seated in the vice resting on a pair of precision ground parallel blocks. I tapped the workpiece down firmly using a copper-faced mallet until the parallel blocks could not be moved by hand, ensuring the part was sitting flat and secure against the floor of the vice.

Manual milling machines feature mechanical clearance play, known as backlash, between the internal lead-screw and the nut assembly inside the table. If you move the table back and forth carelessly, the handwheel dials will give inaccurate position readings.

To eliminate coordinate tracking errors when cutting the plate edges to width, I made sure all table movements were executed from a single direction of travel. If I overshot a target dimension on the dial, I backed the handwheel off by a full turn and re-approached the number from the original direction. This absorbed the internal play, ensuring the table moved exactly what the dial indicated and keeping the opposing sides of the steel plate perfectly parallel.



Fig 5: vertical mill carefully set to be used for machining.

Pillar Drill: Hole Accuracy and Breakthrough Control.

Component 2 (the rectangular aluminium plate) required high accuracy on the

vertical pillar drill to align multiple parallel holes, deep counterbores, and flush-fitting countersinks. Because aluminium is soft, standard twist drills tend to grab the material aggressively, which can cause hole misalignment or rough finishes if the tools are not controlled properly.



Fig 6 : component benign held tight to ensure accuracy.

Center Accuracy and Wandering.

Standard twist drills lack a sharp cutting point at the very center; instead, they have a flat chisel edge. If you try to drill into a flat metal surface directly, the drill bit will flex and "wander" across the surface, ruining the coordinate placement. To prevent this, I marked out the hole centers accurately using a height gauge and a center punch. I then used a short, rigid centre drill to cut a clean, shallow cone at each location. This starter pocket gave the subsequent twist drills a secure guide path, keeping the hole coordinates accurate to the drawing.

Counterbore and Countersink Depth Control.

The drawing required the counterbores to sit at precise depths to hide socket-head cap screw heads perfectly. Because aluminium cuts quickly, it is easy to plunge a tool too deep.

To prevent this error, I utilized the pillar drill's mechanical depth stop collar. I brought the counterbore cutter down until it just touched the top face of the plate, locked the depth stop rings at the exact required dimension on the spindle scale, and executed the cut. The mechanical stop blocked the lever from plunging any further, ensuring every counterbore depth matched the drawing exactly.

Breakthrough Dynamics.

The most critical part of drilling the main through-holes occurred at the very bottom of the cut. As the drill bit point starts to break through the bottom face of the plate, the metal wall thins out significantly. At this exact millisecond, the drill bit point stops cutting cleanly and begins acting like a coarse screw thread. It grabs the remaining thin skin of metal, trying to pull the workpiece upward and out of the vice, a dangerous workshop reaction known as the "helicopter effect."

To manage this breakthrough hazard, I closely monitored the feed lever depth and significantly reduced downward hand pressure just as the drill point began to exit the bottom face. This gentle, controlled feed rate trimmed away the final edge skin smoothly, preventing the part from lifting and leaving a clean, burr-free exit hole.

METROLOGY, TOLERANCING, AND QUALITY CONTROL.

Drawing Interpretation and Geometric Tolerancing.

To manufacture all three components successfully, I had to interpret engineering drawings detailed to standard BS 8888 technical product specifications. The drawings used a third-angle orthographic projection system, which required me to

mentally visualize three-dimensional shapes from flat front, side, and plan views before cutting any metal.

Understanding geometric dimensioning and tolerancing (GD&T) symbols was critical to ensuring the individual features fit together correctly:

- **Concentricity:** For Component 1 (the turned plumb bob), a concentricity symbol was applied to the outer tapered body relative to the top cylindrical section. This meant the centerline of both features had to line up perfectly. Any error during setup, such as an off-center lathe chuck or a speck of swarf trapped behind the raw bar, would cause "run-out," making the plumb bob wobble when spinning and leading to an instant quality rejection.
- **Parallelism and Perpendicularity:** For Component 3 (the milled steel block), the drawing demanded that the opposing cut faces remain parallel to within 0.05 mm, and the side edges remain strictly perpendicular (at a 90° angle) to the bottom datum face. Hitting this required me to keep the milling vice jaws meticulously clean and use precision parallel blocks to maintain true vertical and horizontal alignment.
- **True Positional Tolerance:** For Component 2 (the drilled aluminium plate), the positions of the holes, counterbores, and countersinks were defined by exact X and Y coordinates from a master corner datum. Missing these tolerances would mean the mating bolts would not line up during final assembly.

Quantitative Dimensional Variance Analysis (Data Table).

To verify if my parts met industrial standards, I conducted a formal inspection using calibrated inspection tools. The table below represents the absolute inspection record of critical dimensions across all three manufactured components:

Component And Target Feature	Drawing Nominal Specification	Measured Dimension	Absolute Dimensional Error	Inspection Metrology Tool Used	Industrial Acceptance Status
Comp 1: Plumb Bob Max OD	∅ 25.00 mm ± 0.05.	24.97 mm	-0.03 mm	Calibrated External Micrometer	PASS
Comp 1: Taper Axial Length	50.0 mm ± 0.10	50.06 mm	+0.06 mm	Vernier Depth Gauge	PASS
Comp 2: Counterbore Depth	6.0 mm ± 0.10	5.95 mm	-0.05 mm	Digital Vernier Caliper	PASS
Comp 2: Hole	20.0 mm ±	20.14 mm	+0.14 mm	Digital	FAIL (Out of

1 Center (X)	0.10			Readout (DRO) / Caliper	Spec)
Comp 3: Milled Block Width	45.0 mm ± 0.05	44.98 mm	-0.02 mm	Calibrated External Micrometer	PASS

Detailed Error Source Evaluation.

An honest, professional analysis of this data highlights a critical lesson regarding tool control on the shop floor. While four out of the five critical checked dimensions passed inspection, the coordinate position for Hole 1 on Component 2 (the drilled aluminium plate) failed inspection at + 0.14mm over the allowable limit.

Tracing back the root cause of this out-of-spec dimension revealed an error in my setup behavior. On this specific hole location, I inadvertently skipped the center-drilling step and attempted to plunge the main twist drill bit directly onto the flat aluminium face.

Because standard twist drills feature a blunt chisel edge rather than a sharp point, the tool tip flexed and skated slightly across the slick aluminium surface before biting into the metal. This slight tool deflection shifted the final axis of the hole by 0.14mm.

In a commercial setting, this error would render the plate scrap, as it would cause the mating socket-head screw to bind during assembly. This taught me a lasting lesson: skipping foundational steps like center-drilling to save time always backfires and compromises part quality.



Fig 7 : finished component 2.

PROFESSIONAL BEHAVIOURS AND PERSONAL REFLECTION.

Proactive Problem Solving and Tool Management.

A major requirement of the assignment brief is demonstrating how good communication and proactive problem-solving result in a worn tool being replaced, keeping workshop activities running smoothly. This exact scenario occurred during a roughing pass across the EN8 carbon steel plate on the vertical milling machine.

While executing a longitudinal shoulder cut, I noticed an unexpected change in the machine's operational feedback. A sudden, high-frequency squealing noise developed, accompanied by an increase in structure-borne vibration through the milling machine knee.

Instead of ignoring these signs and letting the machine run, I took immediate, proactive problem-solving action:

1. I hit the feed stop to isolate the machine axes safely.
2. Once the spindle came to a complete rest, I conducted a visual inspection of the indexable end mill cutter.

3. Using a 10× optical loupe, I checked the tool geometry and discovered a chipped flank land and clear crater wear on two of the primary cutting tip inserts. This wear explained the rubbing sound and increased vibration.

I practiced good engineering communication by immediately reporting the tool condition to the workshop supervisor. I explained that continuing the cut with chipped inserts would ruin the steel surface finish, accelerate tool failure, and potentially damage the expensive milling machine spindle due to high cutting forces.

After getting approval to change the inserts, I took the initiative to fix the issue properly:

- I used a torque wrench to remove the worn carbide inserts and fitted brand-new ones into the cutter body.
- Because the new inserts had slightly different physical dimensions than the worn ones, I manually adjusted the machine's Z-axis tool height offsets.
- I restarted the spindle and resumed the profiling pass.

The high-frequency squeal disappeared completely, and the machine returned to a smooth, stable hum. This process intervention saved the component from becoming scrap, protected the workshop equipment, and proved I could manage common production problems using sound engineering judgment.

Independence and Professional Growth.

The practical tasks in Assignment 2 marked a significant step forward in my transition from a trainee who needs constant supervision to an independent professional. On the shop floor, demonstrating independence means taking full responsibility for setting up, adjusting, and running machinery safely without needing a supervisor to double-check every movement.

I demonstrated this professional behavior across several critical setup phases during the manufacturing runs:

- **Lathe Setup Independence:** When preparing to turn the conical plumb bob on the manual lathe, I took responsibility for calculating and selecting my own spindle speeds and feed rates based on the material properties of 6082-T6 aluminium. I manually set up the gear levers on the headstock and independent compound slide without relying on peer support.
- **Milling Setup Accuracy:** On the vertical mill, I took the initiative to independently set up the Dial Test Indicator (DTI) to clock the vice jaw true. I carefully adjusted the mounting bolts until the vice was square within the required ± 0.02 mm tolerance, ensuring my workholding setup was mathematically sound before starting.

- **Pillar Drill Depth Controls:** On the vertical pillar drill, I took full ownership of setting up the mechanical depth stop collars to control the countersink and counterbore depths on the aluminium plate. Rather than guessing, I used a digital vernier caliper to carefully measure my trial depth, adjusted the locknuts on the spindle sleeve, and locked them down securely.

Reflecting on my performance, taking responsibility for these adjustments shows true professional growth. Instead of waiting for instructions, I analyzed the engineering drawings to BS 8888 standards, set up the appropriate tooling, and adjusted parameters based on cutting physics.

While the failed coordinate position on the drilled plate was an expensive lesson, identifying my own mistake, skipping the center-drilling step, and understanding exactly why the tool wandered demonstrates an analytical engineering mindset.

Learning from these mistakes and knowing how to prevent them next time is a key part of continuous professional development, proving I can meet the rigorous quality and safety standards expected in a modern commercial engineering company.

RECOMMENDATIONS FOR COMMERCIAL OPTIMIZATION. Process Efficiency and Sustainable Manufacturing.

To upgrade these manual setups to a commercial standard, the manufacturing process must be modified to minimize production cycle times and reduce environmental waste. In a competitive engineering environment, time and material waste represent direct financial losses.

Material Segregation and Circularity.

During the practical runs, swarf from the two aluminium components and the single EN8 carbon steel block was collected in the same general base trays. In a commercial facility, this mixing destroys the scrap value of the metal.

- **Action:** I recommend installing dedicated, labeled chip containment trays on the lathe, mill, and pillar drill to separate 6082-T6 aluminium from EN8 carbon steel swarf.
- **Benefit:** Clean, unmixed aluminium swarf can be sold back to material suppliers for recycling at up to 70% of its raw material cost. This supports industrial sustainability targets and creates a second revenue stream for the business.

Fluid Life Management.

The water-mix soluble cutting oil emulsion used across the milling machine and pillar

drill should be routed through a centralized fluid filtration and skimmer system. Instead of dumping old coolant down the drain, a skimmer removes tramp oil and bacteria. This extends the lifespan of the cutting fluid from a few weeks to several months, reducing chemical disposal costs and lowering the workshop's environmental impact.

Tooling and Technological Upgrades.

The manual setups used during the assignment can be modernized with specific hardware upgrades. These changes will eliminate human tracking errors, increase machining accuracy, and speed up setup times.



Fig 8 : tools used for machining are selected to produce the best work.

Manual Centre Lathe: Quick-Change Collet System.

- The Issue: Using a standard manual 3-jaw chuck to hold the soft aluminium bar for the plumb bob requires fitting temporary sheet-metal shims. This step is necessary to prevent the hardened chuck jaws from leaving deep indentations on the soft metal, which adds time to the setup.
- The Upgrade: Replace the manual 3-jaw chuck with a lever-operated quick-change collet chuck system.
- The Benefit: A collet wrap applies uniform radial clamping force around the entire circumference of the aluminium bar. This completely eliminates localized surface marks, removes the need for temporary shims, reduces part loading times by 80%, and improves concentricity.

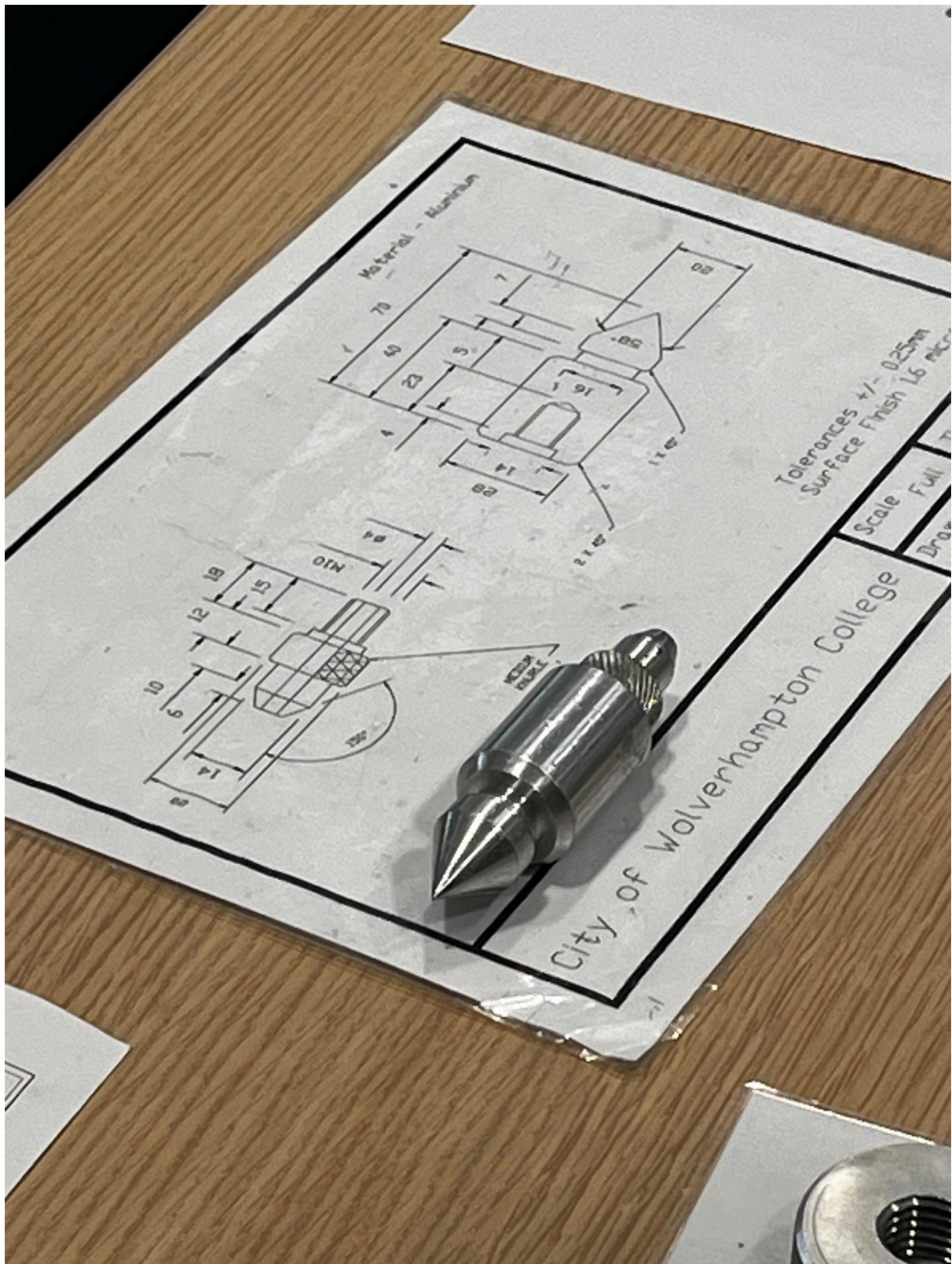


Fig 9: finished and polished component.

Vertical Milling Machine: 3-Axis DRO and Infrared Edge Finder.

- The Issue: Relying on manual handwheel dials requires the operator to constantly calculate and compensate for lead-screw backlash play. If the operator forgets to approach a coordinate from a single direction, the entire part can be ruined.
- The Upgrade: Retrofit the milling table axes with a 3-Axis Digital Readout (DRO) system paired with an infrared edge-finding probe.
- The Benefit: A DRO tracks table movement using glass scales mounted directly to the machine slides, completely bypassing internal lead-screw play. The operator can read actual position data instantly on a digital screen. Combining this with an electronic probe allows the operator to locate the true corner datum of the EN8 steel block in seconds, removing human visual errors and ensuring opposing sides remain parallel.

Vertical Pillar Drill: Precision Coordinate Milling Vice.

- The Issue: Clamping the rectangular aluminium plate in a standard drill vice requires the operator to manually unclamp, move, and re-align the vice for every single hole location. This process can cause tool wander, as seen in the Hole 1 failure.
- The Upgrade: Mount a precision coordinate cross-slide vice securely to the pillar drill table.
- The Benefit: Once the rectangular plate is squared and locked into the vice, the operator can use precise X and Y lead-screws to dial in the coordinates for the pilot holes, counterbores, and countersinks without loosening the main clamp. This keeps the tool aligned with the workpiece datum, ensuring hole coordinates remain accurate and reducing tool deflection.



Fig 10; component benign machined. Precision is key here.

CONCLUSION.

This report demonstrates that manufacturing competency goes beyond simply cutting metal to a drawing. True engineering proficiency requires an integrated understanding of workplace health and safety frameworks, materials technology, workshop metrology practices, and a reflective approach to personal performance.

Reviewing the practical tasks in Assignment 2 highlights several critical areas of growth and technical understanding:

- **Safety Mastery:** Proactively identifying the entanglement hazard on the vertical pillar drill and modifying the process to a 2 mm peck-drilling cycle proved that risk management is an active tool on the shop floor.
- **Technical Knowledge:** Optimizing lathe parameters by increasing spindle speed to 1800 rpm successfully countered the adhesive welding mechanism of Built-Up Edge (BUE), producing a high-quality surface finish on the aluminium plumb bob.
- **Accountability:** Analyzing the out-of-spec coordinate error on the drilled plate taught a valuable lesson. Skipping the center-drilling step allowed the twist drill's chisel edge to wander across the material surface, illustrating why foundational steps must never be skipped to save time.
- **Professional Behavior:** Recognizing a failing tool edge on the milling machine, isolating the equipment safely, communicating with the supervisor, and replacing the inserts proved my transition toward independent professional operation.

By implementing the commercial recommendations outlined in this report, such as material segregation, quick-change collet systems, and digital readouts, a machine shop can achieve higher part accuracy, lower scrap rates, and better production sustainability. These experiences form a strong practical foundation that I will carry forward into the next stage of my professional engineering career.

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